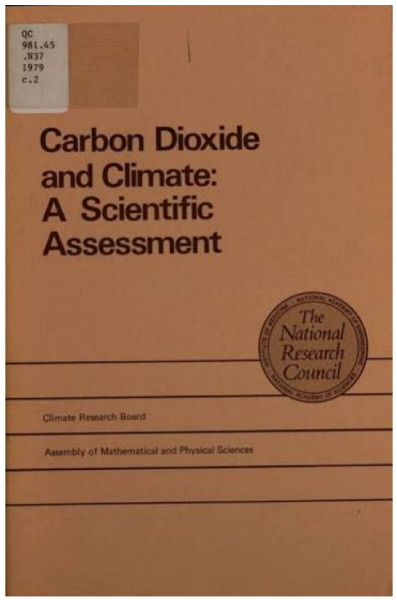
From Charney to CMIP: the evolution and role of Model Intercomparison Projects

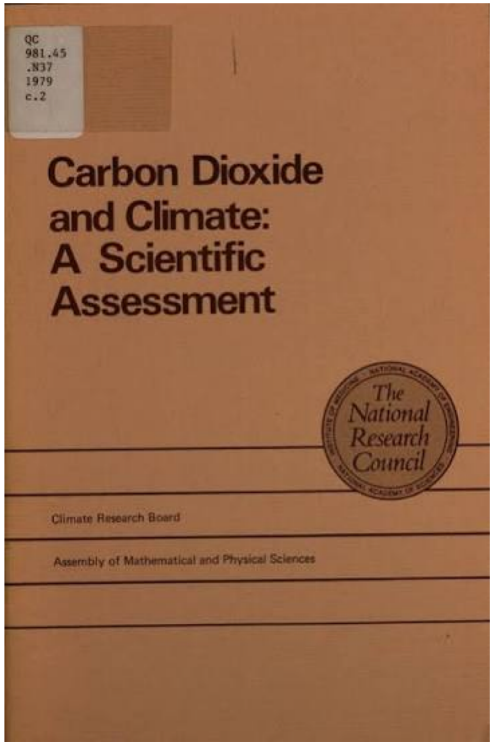


Michael Previdi

Lamont-Doherty Earth Observatory

5 November 2026

CMIP
Coupled Model Intercomparison Project



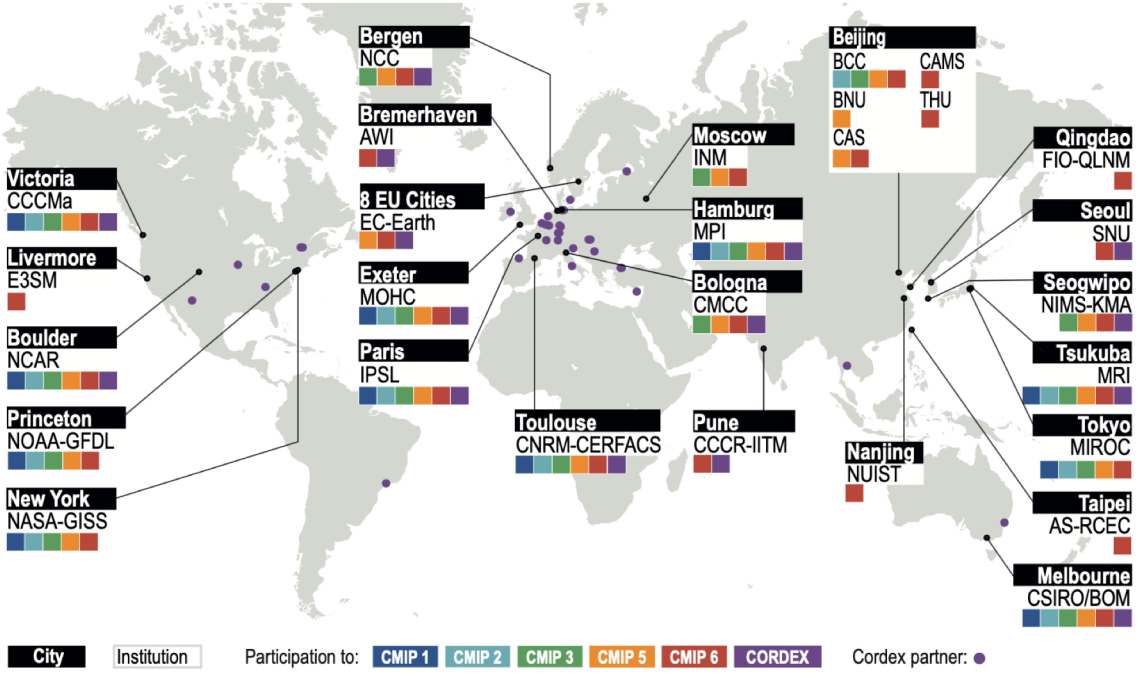
Climate Sensitivity: amount of global-mean surface warming following a doubling of atmospheric CO_2

"As climate modeler Syukuro Manabe of Princeton University tells it, formal assessment of climate sensitivity got off to a shaky start. In the summer of 1979, the late Jule Charney convened a committee of fellow meteorological luminaries on Cape Cod to prepare a report for the National Academy of Sciences on the possible effects of increased amounts of atmospheric CO2 on climate. None of the committee members actually did greenhouse modeling themselves, so Charney called in the only two American researchers modeling greenhouse warming, Manabe and James Hansen of NASA's Goddard Institute for Climate Studies (GISS) in New York City."

"On the first day of deliberations, Manabe told the committee that his model warmed 2°C when CO2 was doubled. The next day Hansen said his model had recently gotten 4°C for a doubling. According to Manabe, Charney chose 0.5°C as a not-unreasonable margin of error, subtracted it from Manabe's number, and added it to Hansen's. Thus was born the 1.5°C-to-4.5°C range of likely climate sensitivity that has appeared in every greenhouse assessment since, including the three by the Intergovernmental Panel on Climate Change (IPCC). More than one researcher at the workshop called Charney's now-enshrined range and its attached best estimate of 3°C so much hand waving."

Kerr (2004)

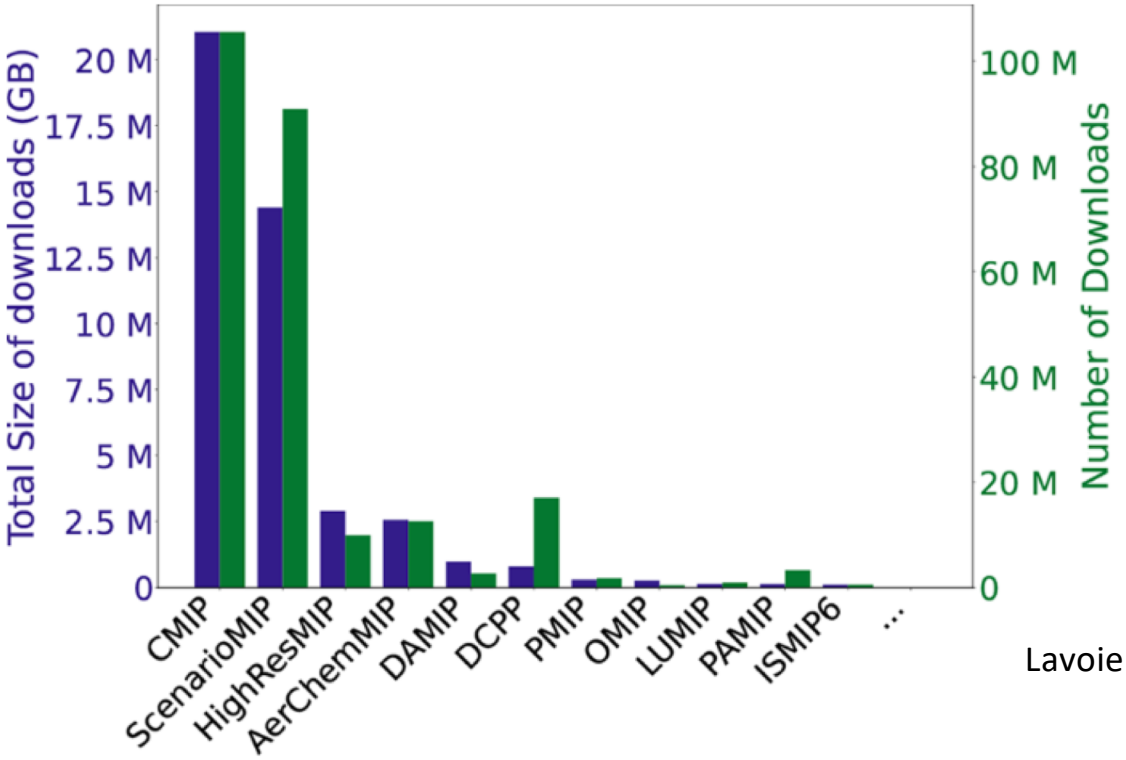
Coupled Model Intercomparison Project



AR6 WG1, Ch. 1

Today climate modeling is a multinational effort involving dozens of Earth System Models, with a large part of this effort organized through the Coupled Model Intercomparison Project (CMIP).

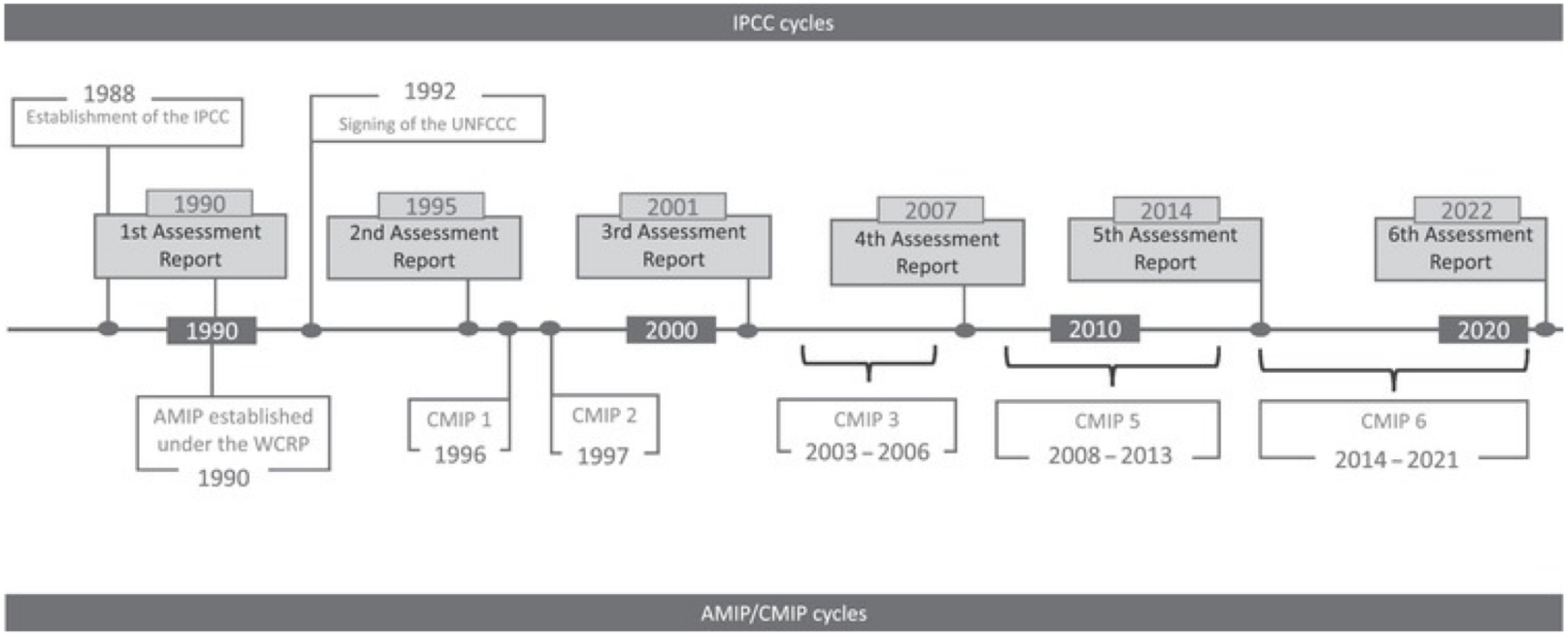
The big data challenge of CMIP



CMIP6 data downloads

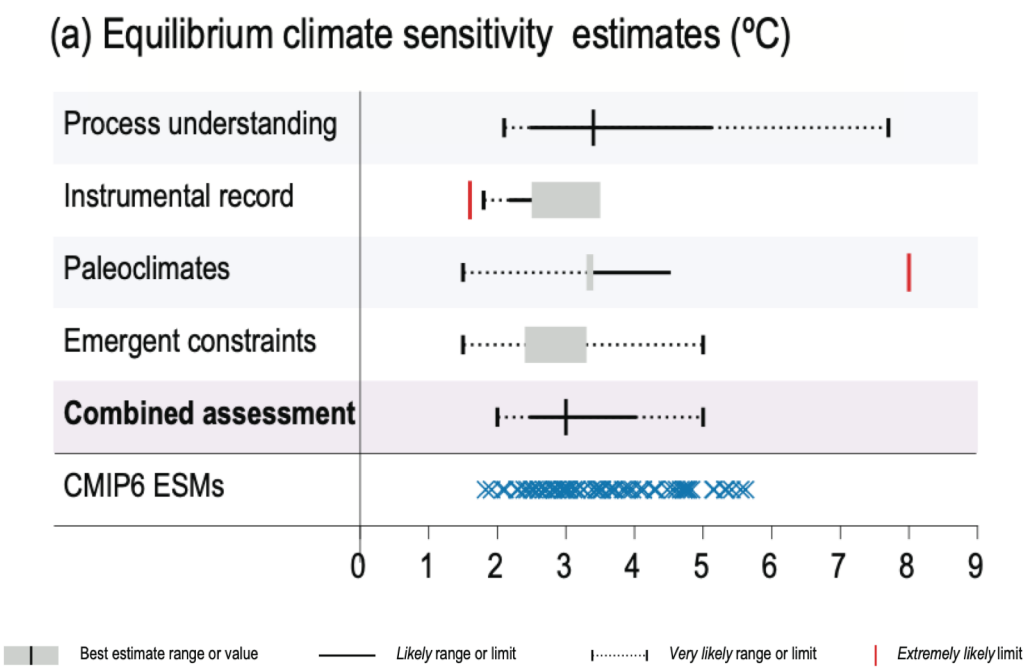
Lavoie et al. (2026)

Role of CMIP in IPCC Assessments



Guillemot (2022)

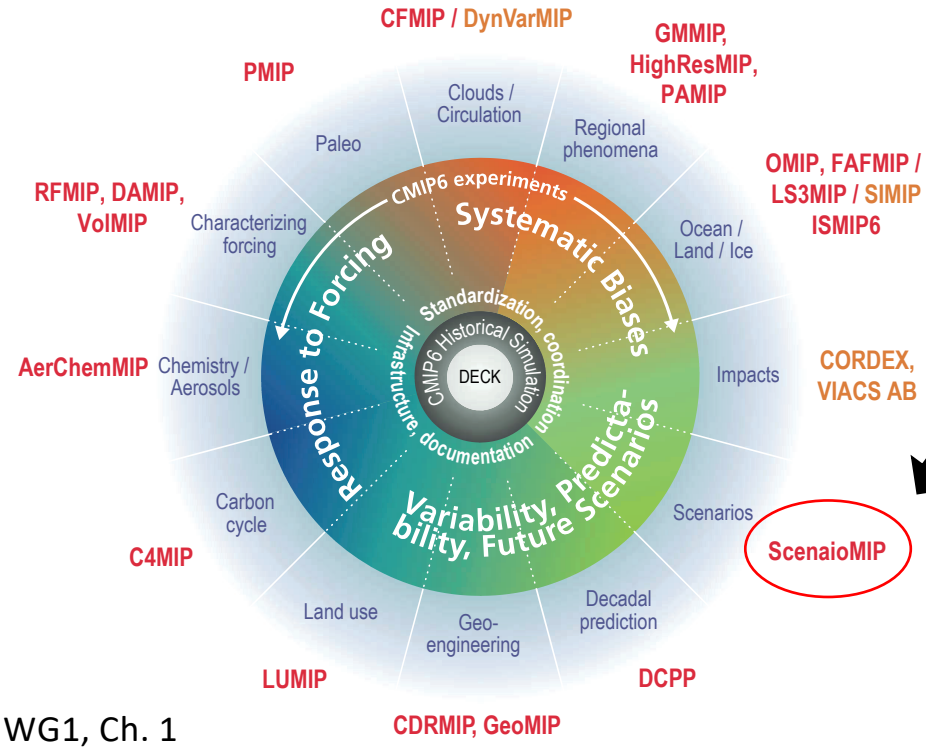
IPCC Assessments don't rely solely on CMIP



IPCC AR6 (2021)

From Figure caption: "CMIP6 model values are not directly used as a line of evidence but presented on the Figure for comparison."

CMIP6 experimental design



ScenarioMIP: includes Earth System Model future climate simulations driven by SSP emissions scenarios

AR6 WG1, Ch. 1

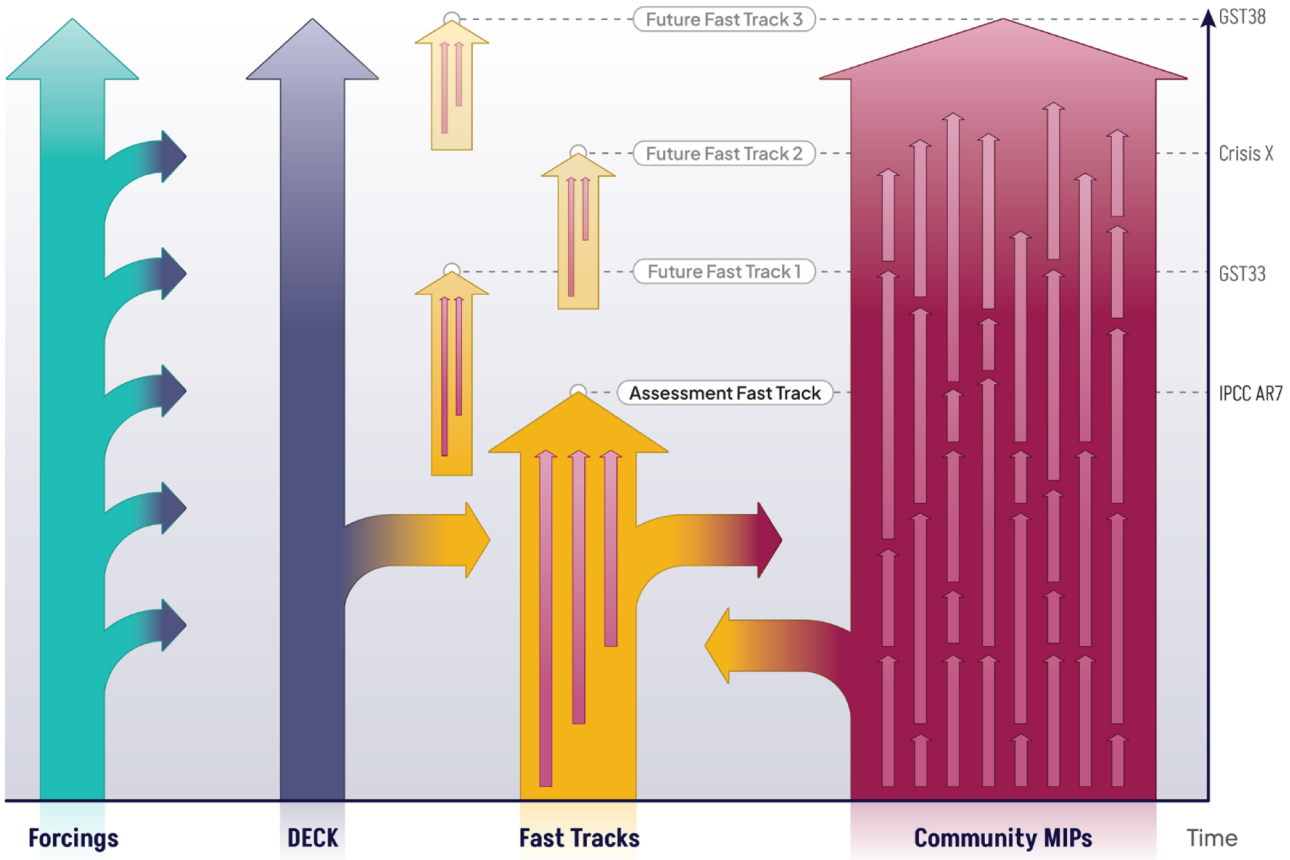
CMIP6 DECK/historical experiments

Experiment short name	CMIP6 label	Experiment description	Forcing methods	Start year	End year	Minimum no. years per simulation	Major purpose
DECK experiments							
AMIP	*amip*	Observed SSTs and SICs prescribed	All; CO_2 concentration prescribed	1979	2014	36	Evaluation, variability
Pre-industrial control	*piControl* or *esm-piControl*	Coupled atmosphere–ocean pre-industrial control	CO_2 concentration prescribed or calculated	n/a	n/a	500	Evaluation, unforced variability
Abrupt quadrupling of CO_2 concentration	*abrupt-4×CO2*	CO_2 abruptly quadrupled and then held constant	CO_2 concentration prescribed	n/a	n/a	150	Climate sensitivity, feedback, fast responses
1 % yr^{-1} CO_2 concentration increase	*1pctCO2*	CO_2 prescribed to increase at 1 % yr^{-1}	CO_2 concentration prescribed	n/a	n/a	150	Climate sensitivity, feedback, idealized benchmark
CMIP6 historical simulation							
Past ~1.5 centuries	*historical* or *esm-hist*	Simulation of the recent past	All; CO_2 concentration prescribed or calculated	1850	2014	165	Evaluation

Eyring et al. (2016)

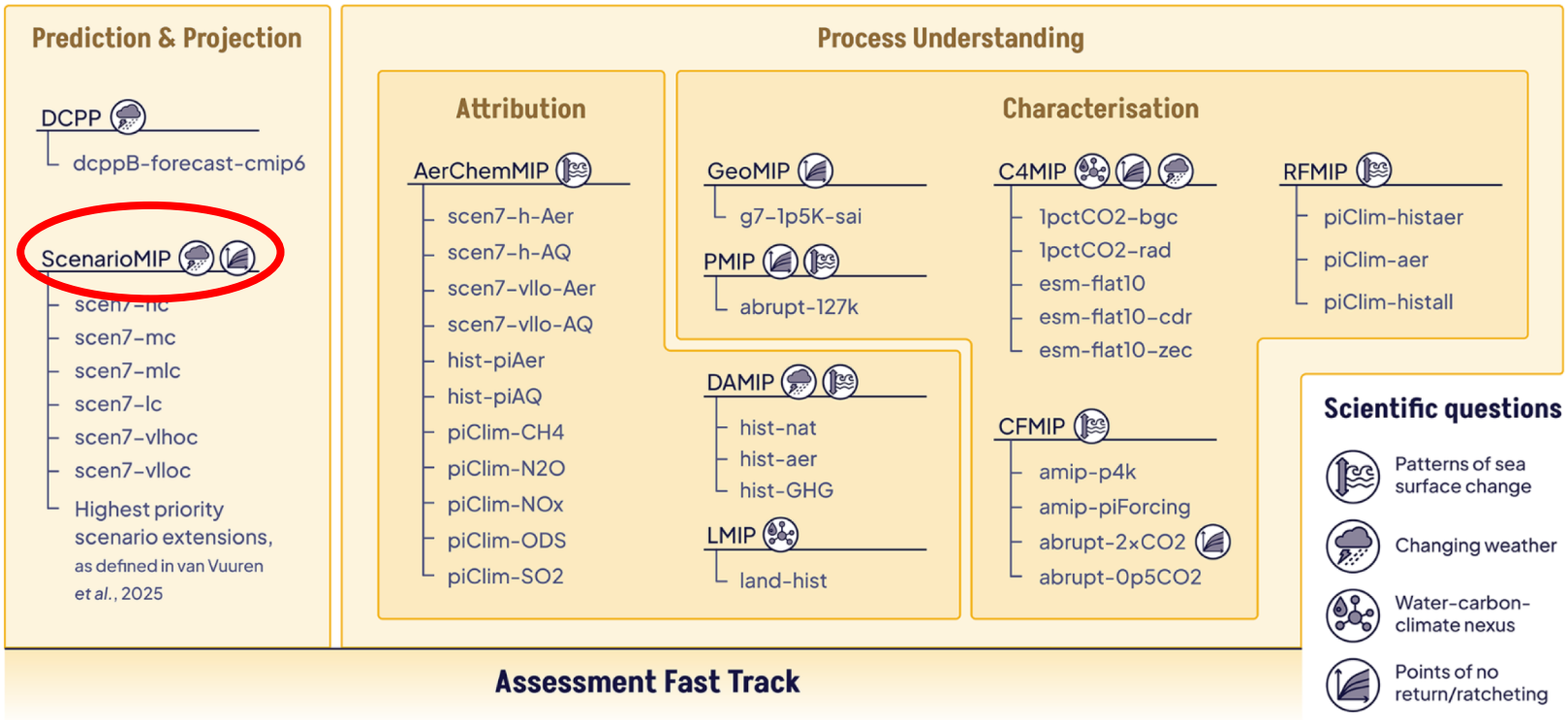
DECK = Diagnostic, Evaluation and Characterization of Klima

CMIP7 experimental design



Dunne et al. (2025)

CMIP7 experimental design



Dunne et al. (2025)

Future scenarios

- A scenario is a description of how the future may develop, based on a coherent and internally consistent set of assumptions about key drivers including demography, economic processes, technological innovation, governance, lifestyles, and relationships among these driving forces.

- Scenarios can also be defined by geophysical driving forces only (e.g., greenhouse gas emissions/concentrations).

- The scenarios used in CMIP6 ScenarioMIP are referred to as the SSP (Shared Socio-economic Pathway) scenarios.

- Creation of the SSP scenarios began with the development of *storylines* (or narratives), which are qualitative descriptions of possible societal futures.

SSP storylines

Summary of SSP narratives.

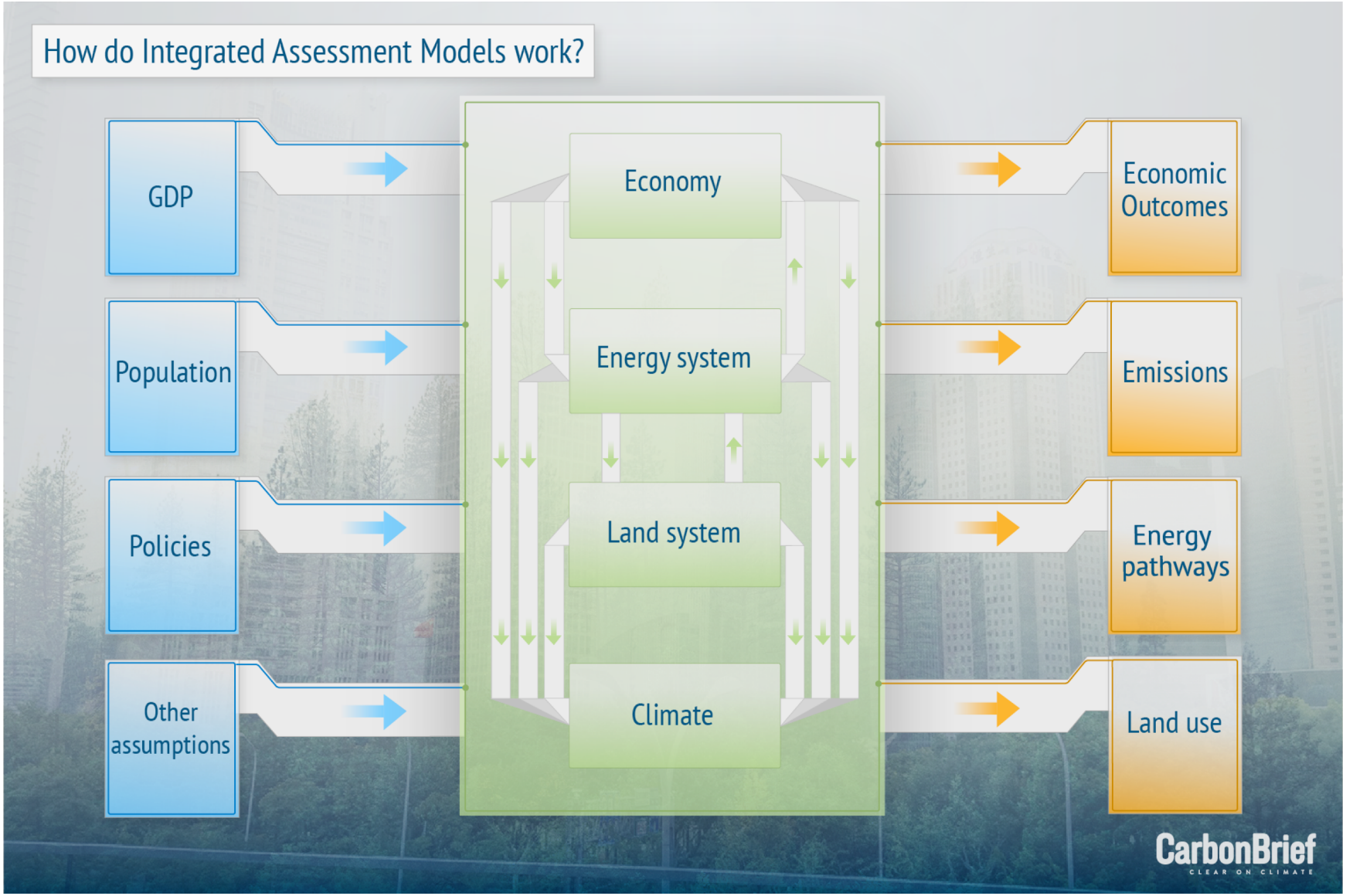
SSP1	Sustainability – Taking the Green Road (Low challenges to mitigation and adaptation) *The world shifts gradually, but pervasively, toward a more sustainable path, emphasizing more inclusive development that respects perceived environmental boundaries. Management of the global commons slowly improves, educational and health investments accelerate the demographic transition, and the emphasis on economic growth shifts toward a broader emphasis on human well-being. Driven by an increasing commitment to achieving development goals, inequality is reduced both across and within countries. Consumption is oriented toward low material growth and lower resource and energy intensity.*
SSP2	Middle of the Road (Medium challenges to mitigation and adaptation) *The world follows a path in which social, economic, and technological trends do not shift markedly from historical patterns. Development and income growth proceeds unevenly, with some countries making relatively good progress while others fall short of expectations. Global and national institutions work toward but make slow progress in achieving sustainable development goals. Environmental systems experience degradation, although there are some improvements and overall the intensity of resource and energy use declines. Global population growth is moderate and levels off in the second half of the century. Income inequality persists or improves only slowly and challenges to reducing vulnerability to societal and environmental changes remain.*
SSP3	Regional Rivalry – A Rocky Road (High challenges to mitigation and adaptation) *A resurgent nationalism, concerns about competitiveness and security, and regional conflicts push countries to increasingly focus on domestic or, at most, regional issues. Policies shift over time to become increasingly oriented toward national and regional security issues. Countries focus on achieving energy and food security goals within their own regions at the expense of broader-based development. Investments in education and technological development decline. Economic development is slow, consumption is material-intensive, and inequalities persist or worsen over time. Population growth is low in industrialized and high in developing countries. A low international priority for addressing environmental concerns leads to strong environmental degradation in some regions.*
SSP4	Inequality – A Road Divided (Low challenges to mitigation, high challenges to adaptation) *Highly unequal investments in human capital, combined with increasing disparities in economic opportunity and political power, lead to increasing inequalities and stratification both across and within countries. Over time, a gap widens between an internationally-connected society that contributes to knowledge- and capital-intensive sectors of the global economy, and a fragmented collection of lower-income, poorly educated societies that work in a labor intensive, low-tech economy. Social cohesion degrades and conflict and unrest become increasingly common. Technology development is high in the high-tech economy and sectors. The globally connected energy sector diversifies, with investments in both carbon-intensive fuels like coal and unconventional oil, but also low-carbon energy sources. Environmental policies focus on local issues around middle and high income areas.*
SSP5	Fossil-fueled Development – Taking the Highway (High challenges to mitigation, low challenges to adaptation) *This world places increasing faith in competitive markets, innovation and participatory societies to produce rapid technological progress and development of human capital as the path to sustainable development. Global markets are increasingly integrated. There are also strong investments in health, education, and institutions to enhance human and social capital. At the same time, the push for economic and social development is coupled with the exploitation of abundant fossil fuel resources and the adoption of resource and energy intensive lifestyles around the world. All these factors lead to rapid growth of the global economy, while global population peaks and declines in the 21st century. Local environmental problems like air pollution are successfully managed. There is faith in the ability to effectively manage social and ecological systems, including by geo-engineering if necessary.*

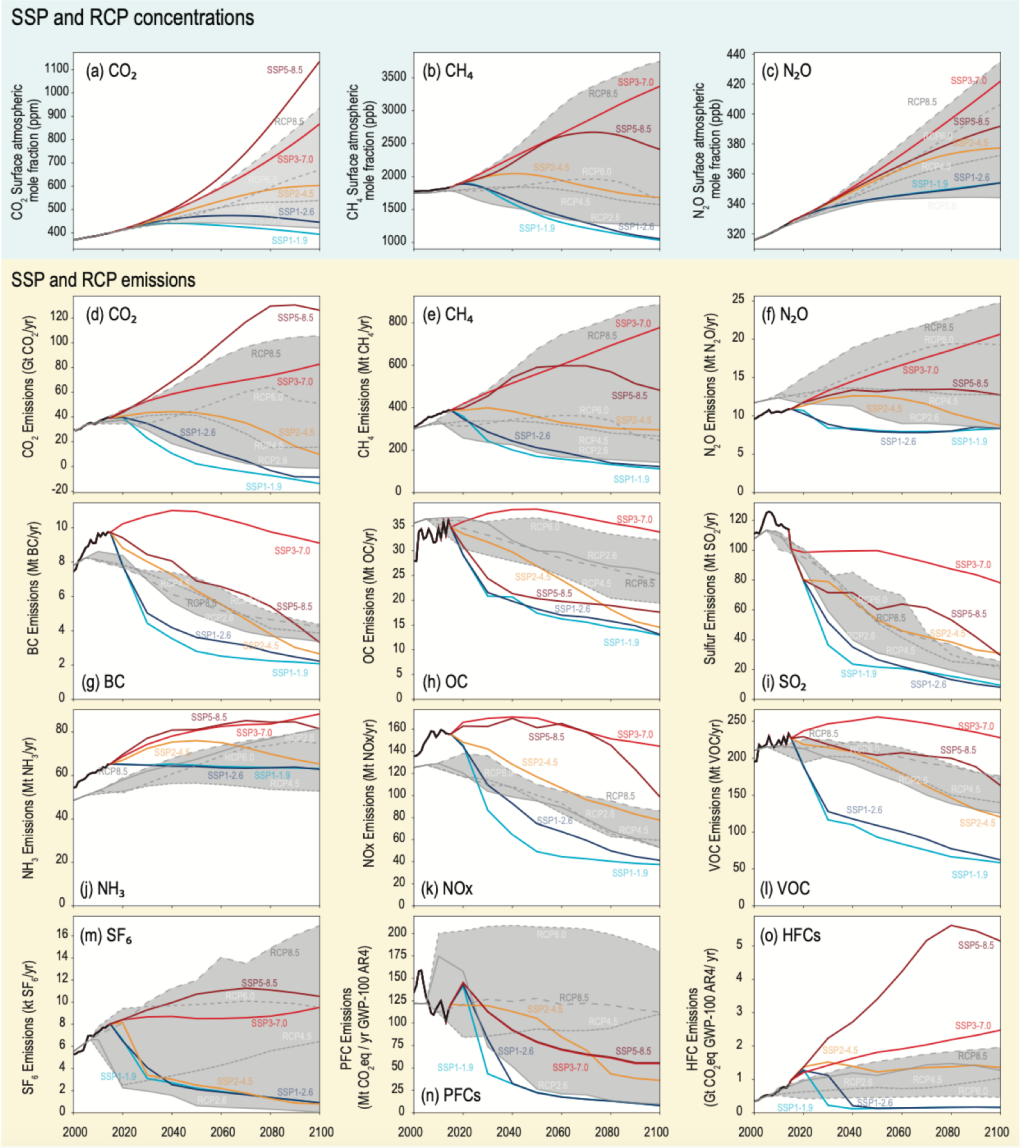
Riahi et al. (2017)

Emissions scenarios

- SSP storylines are used as input to integrated assessment models (IAMs), which translate the associated societal assumptions into scenarios of energy use, air pollution control, land use, and greenhouse gas emissions.

- These emissions scenarios produced by IAMs are then used to drive global climate model simulations in order to project possible future climates (in CMIP6/7, as part of ScenarioMIP).


How do Integrated Assessment Models work?
GDP
Population
Policies
Other assumptions
Economy
Energy system
Land system
Climate
Economic Outcomes
Emissions
Energy pathways
Land use
CarbonBrief
CLEAR ON CLIMATE



SSP emissions scenarios

AR6 uses a core set of five illustrative SSP scenarios: SSP1-1.9, SSP1-2.6, SSP2-4.5, SSP3-7.0 and SSP5-8.5.

The first number in each scenario name refers to an SSP storyline, while the second number refers to the approximate **radiative forcing** in the year 2100.

AR6 WG1, Ch. 1

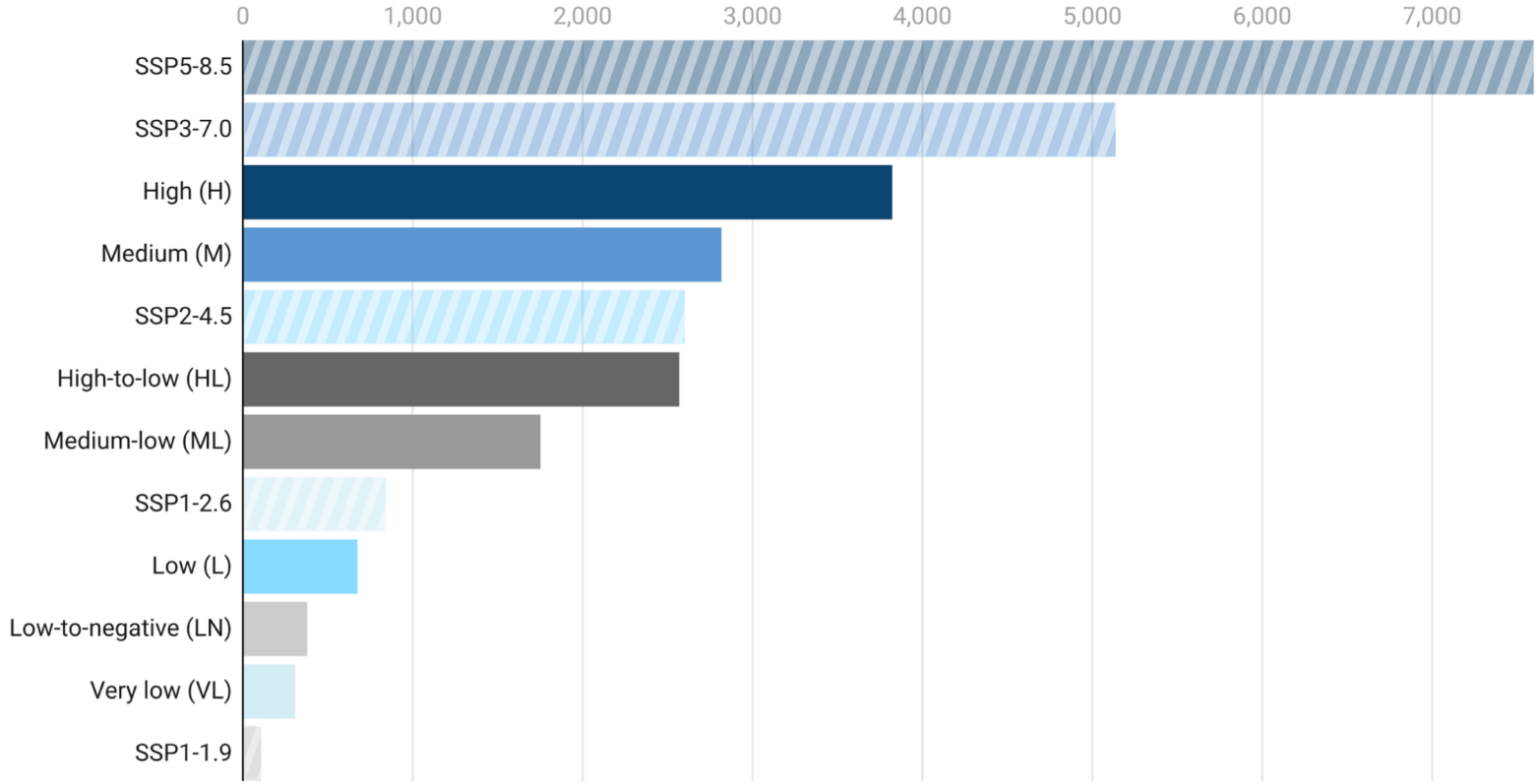
CMIP7 emissions scenarios

Scenario	Marker IAM	Underlying SSP	Emissions pathway	Net CO2 in 2100 (GtCO2/yr)	Cumulative CO2, 2024-2100 (GtCO2)	Warming in 2100 (C vs 1850-1900)
High (H)	GCAM 8s	SSP3	Emissions as high as plausible with policy rollback	55	3,820	3.3 (2.6-4.4)
High-to-low (HL)	WITCH 6.0	SSP5	High to mid-century, then net-zero CO2 by 2100	-1	2,566	2.8 (2.1-4.0)
Medium (M)	IMAGE 3.4	SSP2	Current policies frozen at 2025 levels	34	2,814	2.9 (2.2-3.9)
Medium-low (ML)	COFFEE 1.6	SSP2	Medium until 2040, then decline to net-zero CO2 by 2100	-9	1,757	2.3 (1.7-3.3)
Low (L)	MESSAGEix-GLOBIOM 2.1	SSP2	Aims to keep warming likely below 2C	-9	673	1.8 (1.3-2.7)
Very-low (VL)	REMIND-MAgPIE 3.5-4.11	SSP1	1.5C with as little overshoot as plausible	-6	310	1.6 (1.1-2.5)
Low-to-negative (LN)	AIM 3.0	SSP2	1.5C with higher overshoot, then net-negative greenhouse gases	-25	384	1.7 (1.2-2.5)

Source: Carbon Brief

The CMIP7 scenarios feature much lower emissions at the high end

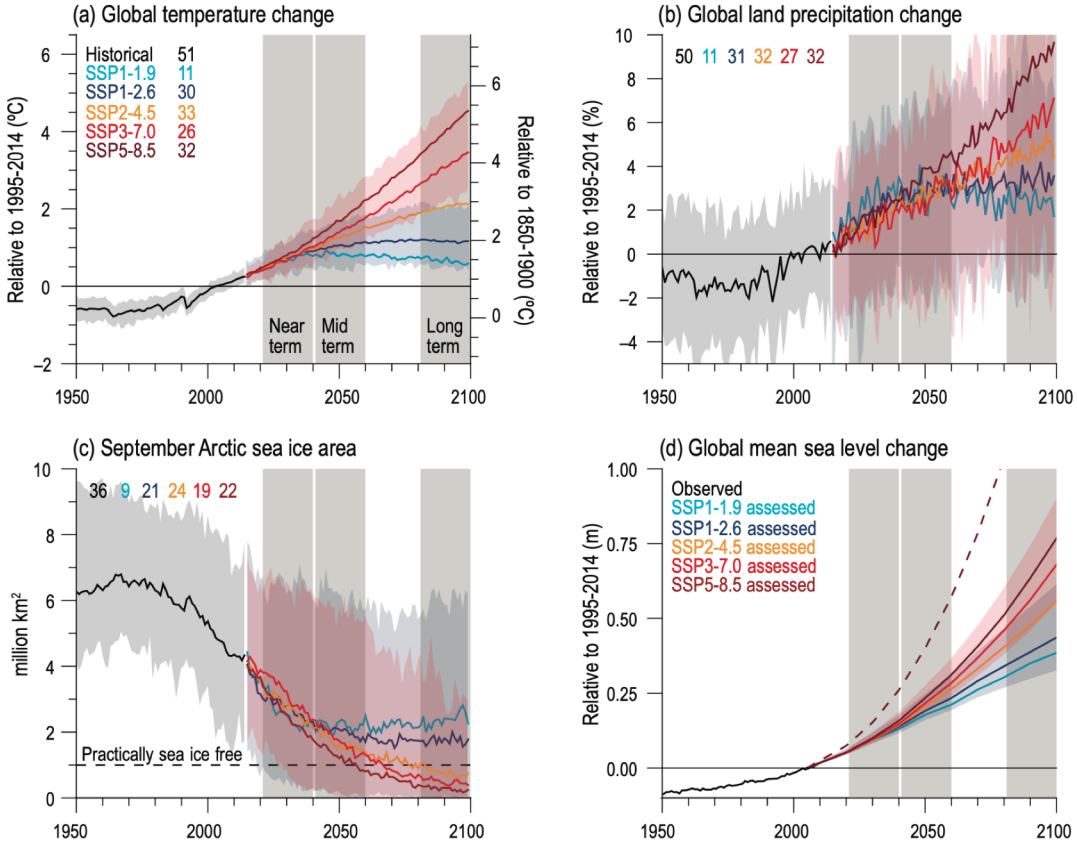
Cumulative net CO2 emissions, 2024-2100, GtCO2



Source: CMIP7 ScenarioMIP marker database (van Vuuren et al. 2026); RCMIP v5.1; analysis by Carbon Brief

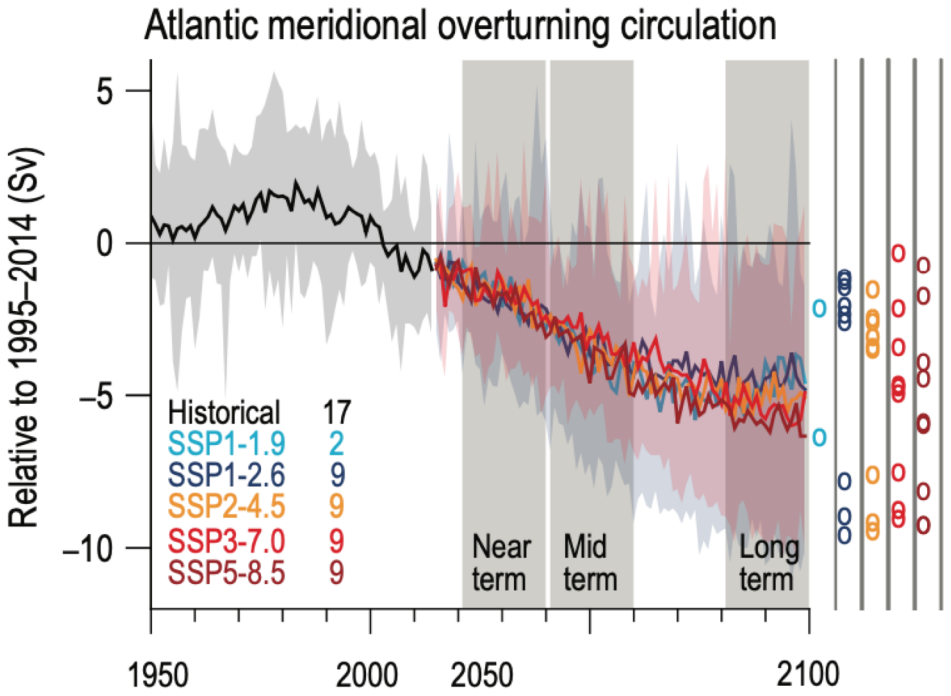
CarbonBrief

Future climate projections from CMIP6 ScenarioMIP



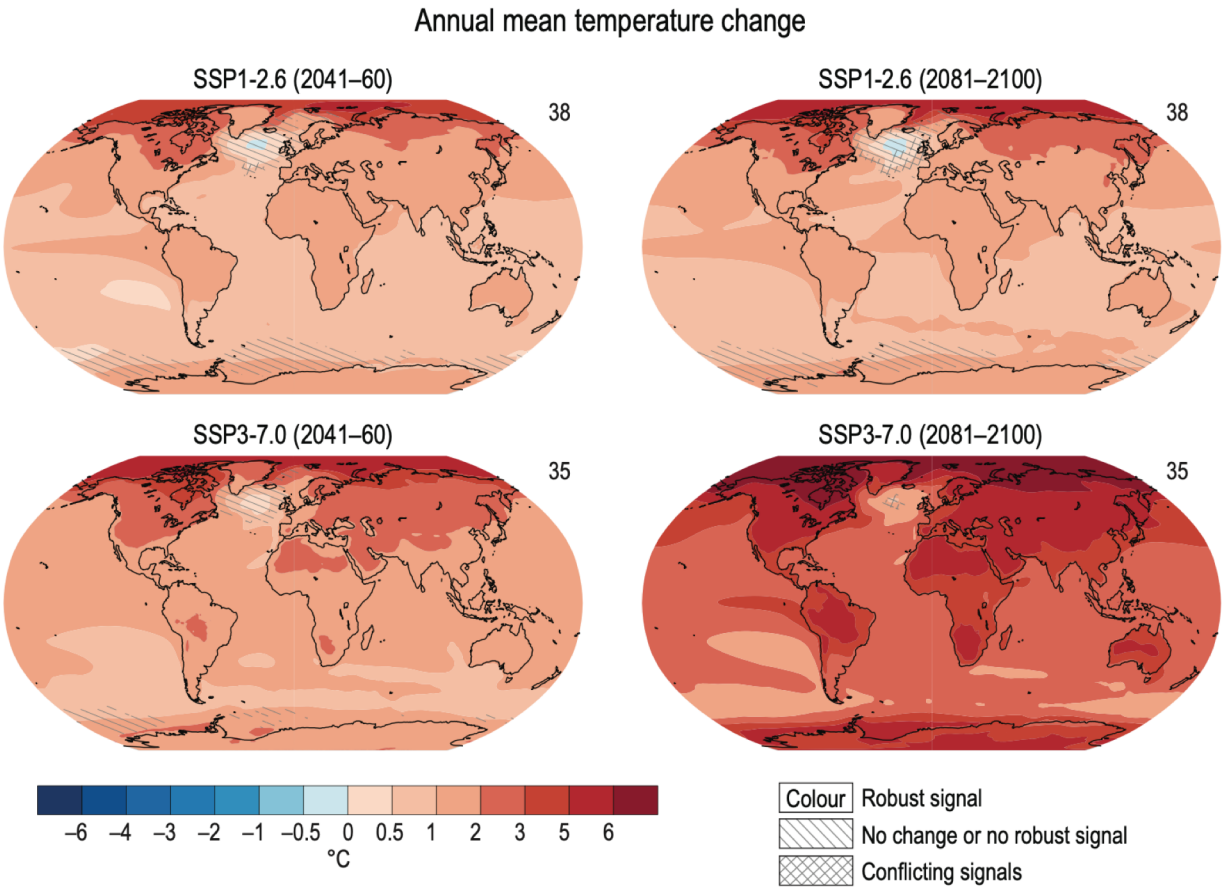
AR6 WG1, Ch. 4

Future climate projections from CMIP6 ScenarioMIP



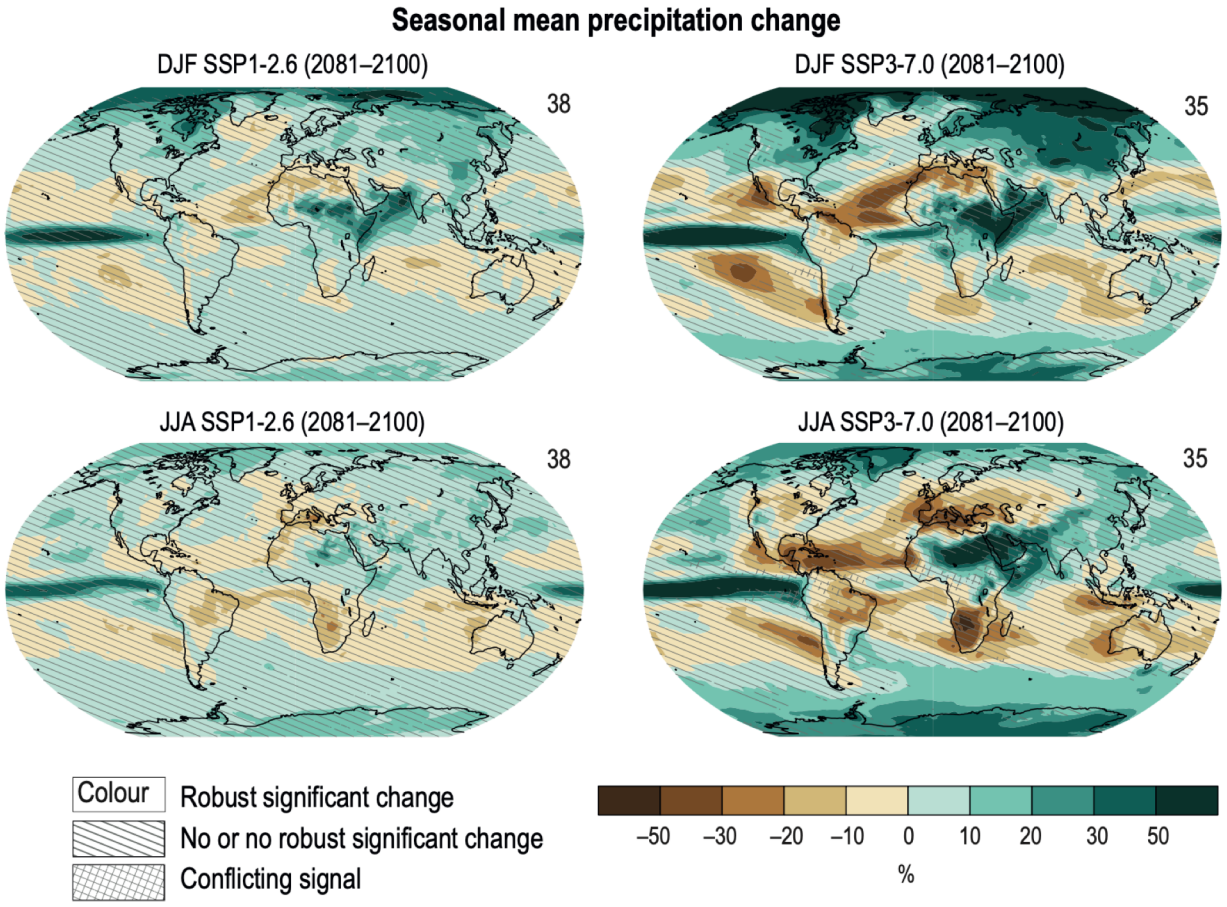
AR6 WG1, Ch. 4

Future climate projections from CMIP6 ScenarioMIP



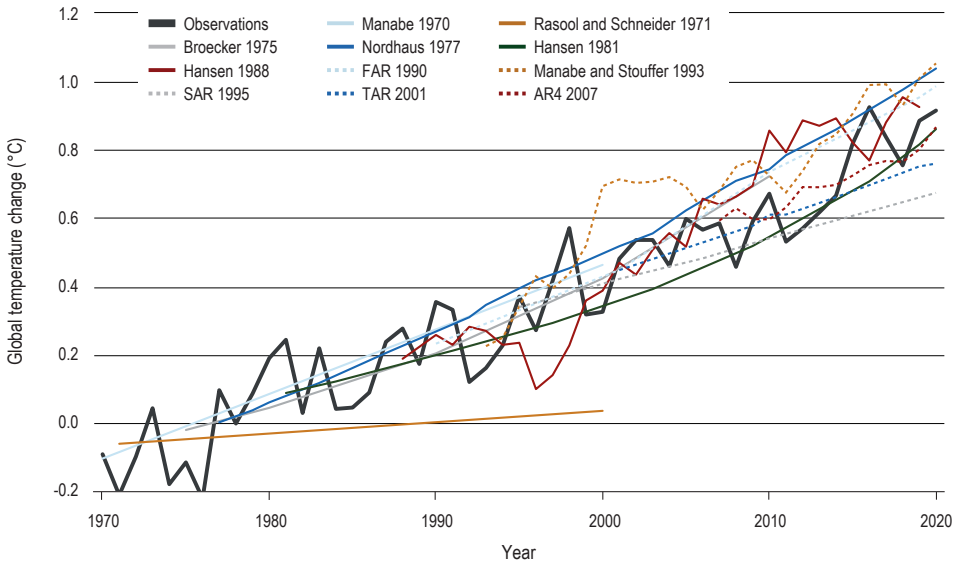
AR6 WG1, Ch. 4

Future climate projections from CMIP6 ScenarioMIP



AR6 WG1, Ch. 4

Should we trust model projections?



Earlier-generation climate model projections of global surface temperature change compared to subsequent observations

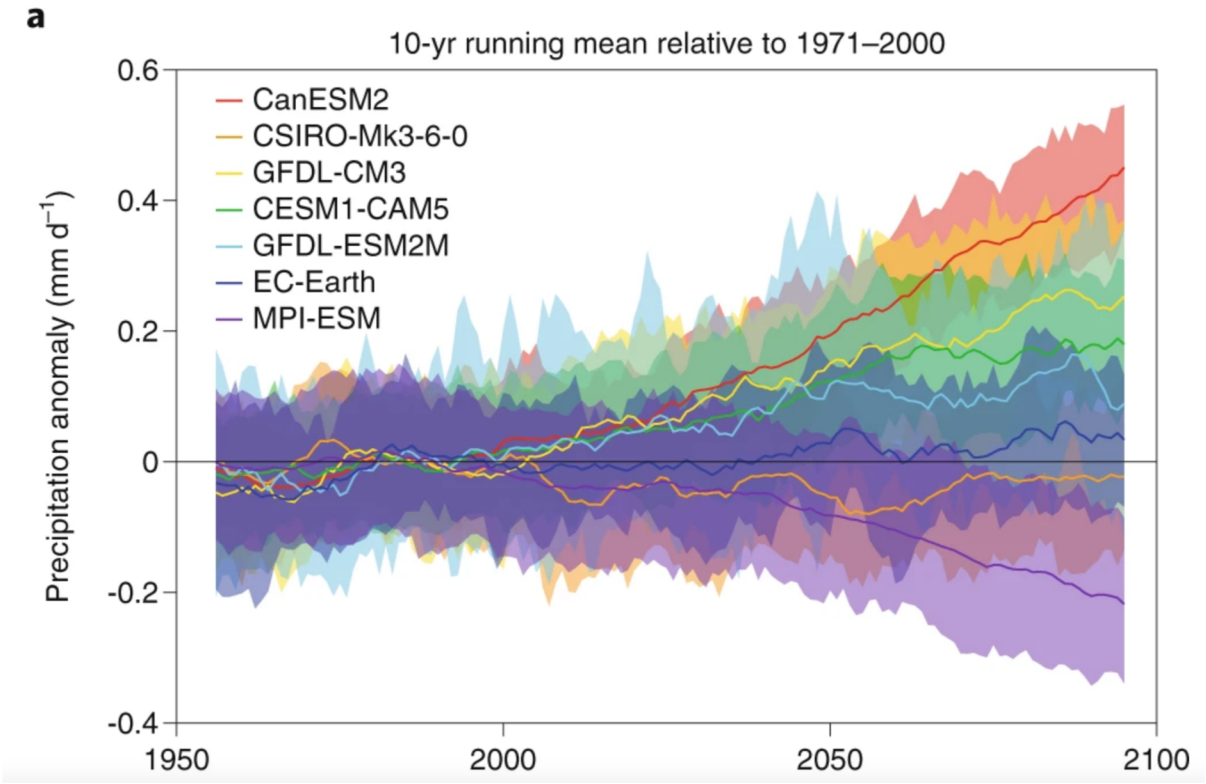
AR6 WG1, Ch. 1

Sources of uncertainty in future climate projections

- Despite evidence of model skill (and increasing skill over time) in simulating many aspects of climate and climate change, future climate projections will always be uncertain. There are 3 main sources of uncertainty:

1. **Scenario uncertainty:** reflects fundamental uncertainty in future human behavior, and associated uncertainty in energy use, greenhouse gas emissions, and other climate change drivers
2. **Model uncertainty:** imperfect models (e.g., missing or parameterized processes) make the climate change response to a given set of drivers uncertain
3. **Internal variability:** uncertainty in the future climate trajectory associated with random, chaotic (unforced) variations in the weather and climate

Sources of uncertainty in future climate projections



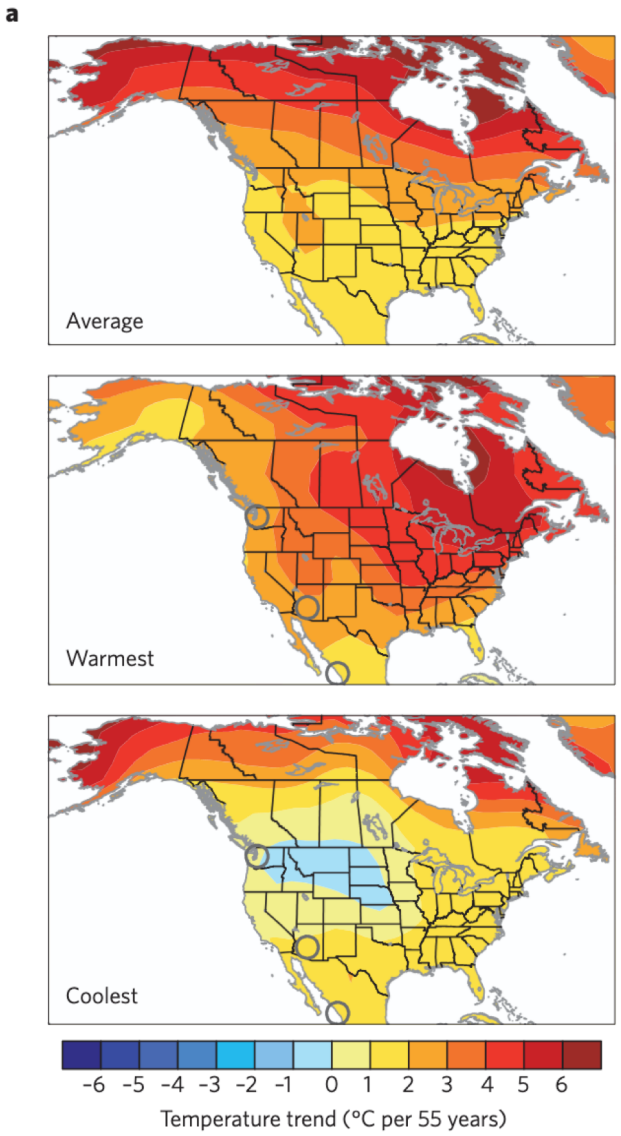
Upper Colorado River Basin annual precipitation

Colored lines = model ensemble means (forced response; difference reflects model uncertainty)

Shading = ensemble spread (reflects uncertainty due to internal variability)

Deser et al. (2020)

Deser et al. (2012): "Communication of the role of natural variability in future North American climate"

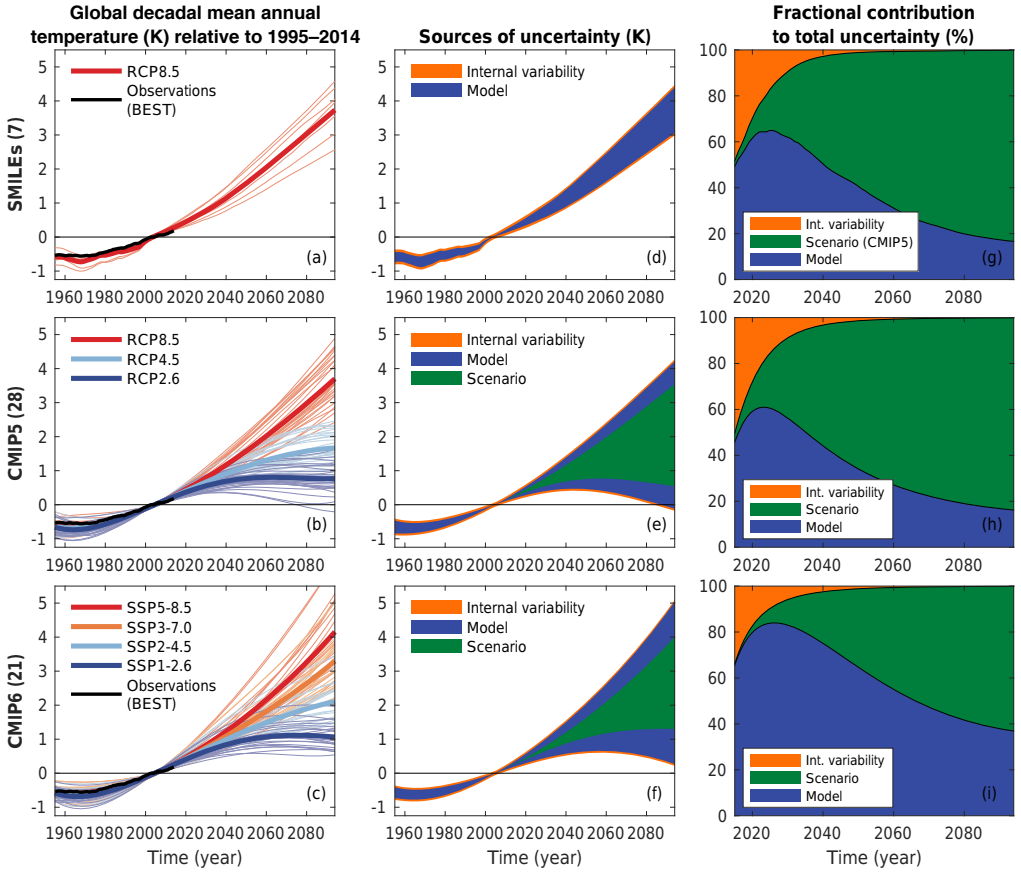


December–January–February surface air temperature trends during 2005–2060 as simulated by a single climate model following a single emissions scenario

Top panel: 40-member ensemble mean (forced response)

Middle and bottom panels: Ensemble members with the warmest and coolest trends for the contiguous United States as a whole

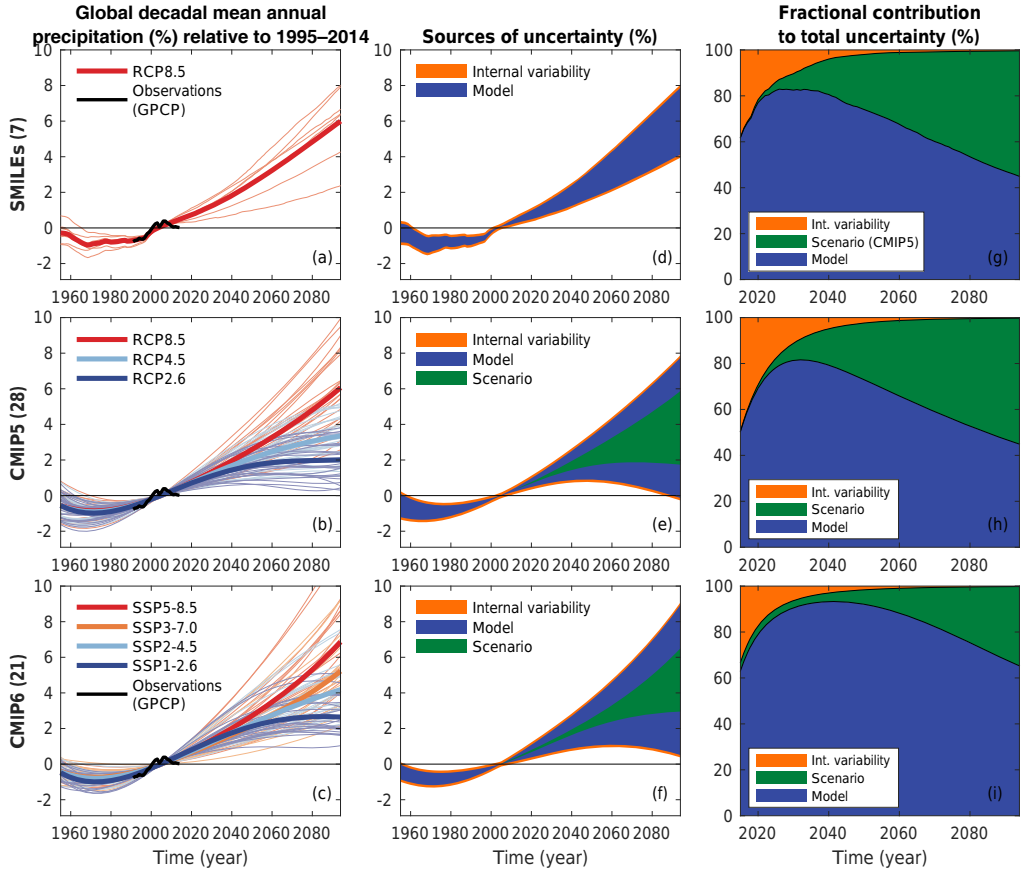
Sources of uncertainty in future climate projections



SMILEs = single-model initial-condition large ensembles

Lehner et al. (2020)

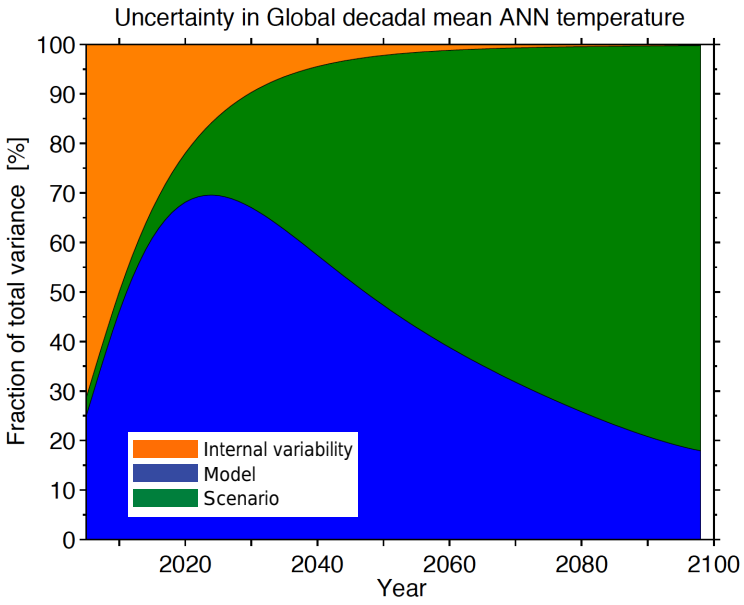
Sources of uncertainty in future climate projections

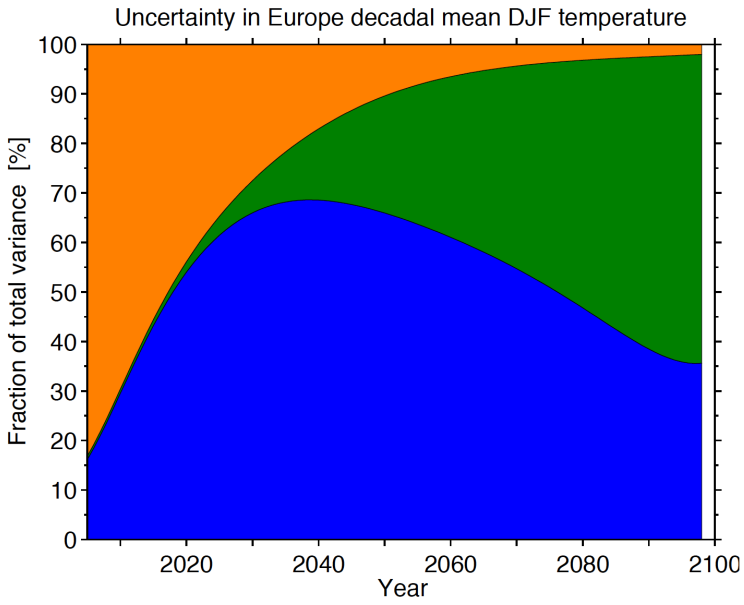


SMILEs = single-model initial-condition large ensembles

Lehner et al. (2020)

Sources of uncertainty in future climate projections





IPCC AR5 (2013)

Internal variability becomes increasingly important on smaller spatiotemporal scales.

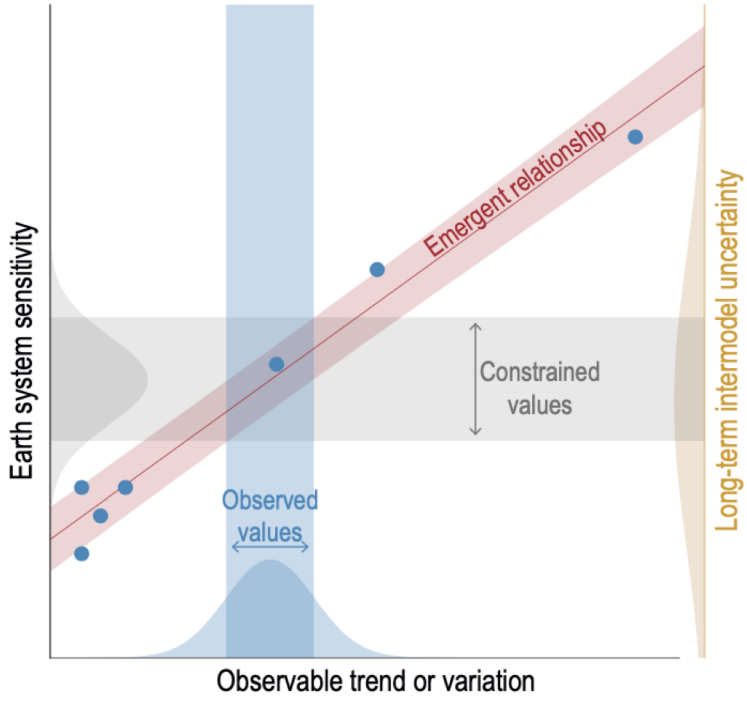
Prospects for reducing uncertainty?

Some amount of uncertainty in future climate projections from all 3 sources is inevitable. However:

1. **Scenario uncertainty:** some future emissions scenarios may be more plausible than others

2. **Model uncertainty:** some models may be more credible than others (model weighting?); emergent constraints (see right) offer the possibility of narrowing model uncertainty

3. **Internal variability:** initialized decadal predictions may reduce uncertainty due to internal variability in the near term



AR6 WG1, Ch. 1

Thank You!

Questions?